DR. FAKHRA GHAFOOR PHYSICS FACULTY | RESEARCHER

Visiting Faculty Candidate | Computational Physics | Condensed Matter Physics | Materials Science

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ACADEMIC PROFILE

Physics educator, researcher, and community builder with teaching experience spanning undergraduate to graduate-level Physics. Experienced in university teaching, academic coordination, assessment, mentoring, computational research, and inclusive academic engagement.

My research combines first-principles computational physics with materials modelling, with particular emphasis on DFT and time-dependent density functional theory (TDDFT) dynamical studies of 2D materials, MXenes, perovskites, bulk and functional materials. Research directions include linear and nonlinear optics, next-generation energy solutions, infrared sensing devices, and computational materials design.

TEACHING & ACADEMIC EXPERIENCE

Lecturer in Physics

University of Wah

2013–2018 Wah Cantt, Pakistan

- Taught university-level undergraduate physics courses with emphasis on conceptual understanding, mathematical derivations, numerical problem solving, and laboratory-linked applications.
- Delivered lectures, tutorials, and structured revision sessions across core areas of physics.
- Prepared course material, assignments, quizzes, examinations, and student feedback aligned with course learning outcomes.
- Contributed to academic coordination and departmental activities, including student affairs and programme-related responsibilities.
- Supported quality assurance and academic documentation, including departmental programme and self-assessment activities.

PhD Researcher / Scholar

Pakistan Institute of Engineering and Applied Sciences (PIEAS)

2018–2025 Islamabad, Pakistan

- Conducted research in computational condensed matter physics and first-principles materials modelling.
- Investigated structural stability, electronic structure, charge transfer, optical response, and thermoelectric properties of two-dimensional materials and heterostructures.
- Developed computational workflows for DFT-based simulations, post-processing, data analysis, scientific visualization, and publication preparation.
- Collaborated on interdisciplinary research involving MXenes, semiconductor heterostructures, perovskites, and energy-related materials.

EDUCATION

PhD in Physics, PIEAS, Islamabad
2018–2025

- **CGPA:** 4.00 / 4.00
- **HEC Scholarship Awardee** during PhD
- **Thesis:** *First Principles Calculations of Zirconium Based MXene Interfaces*
- **Research Area:** Computational Condensed Matter Physics
- **Research Focus:** First-principles modelling and simulation of 2D materials, MXenes, perovskites, bulk materials, linear and nonlinear optical materials, thermoelectric materials, and infrared-responsive materials, with particular emphasis on DFT and TDDFT dynamical studies, optical response, nonlinear optical phenomena, infrared sensing, and next-generation energy solutions.

MS / MPhil in Physics, Quaid-i-Azam University, Islamabad
2011–2013

- **CGPA:** 3.50 / 4.00
- **Scholarships:** Merit Scholarship; Research Scholarship from the World Federation of Scientists (WFS), Geneva, Switzerland.
- **Research Field:** Experimental High Energy Physics
- **Research Topic:** *Study of Single Top Quark Mass Measurement at Tevatron and LHC Centre of Mass Energies*
- Research involved Monte Carlo event generation, data analysis, and studies of single-top production channels.

BS in Physics, University of Wah
2005–2009

- **CGPA:** 3.81 / 4.00
- **Position:** 2nd position holder in the University
- **Final Year Project:** *Optimizing the Efficiency of Solar Power Systems*

SELECTED PUBLICATIONS

- Fizza Siddique, M. Naeem, **F. Ghafoor**, Li. Defect-engineered ultrathin $ZnIn_2S_4$ for efficient photocatalytic hydrogen production. *Journal of Alloys and Compounds*, under review (2026).
- **F. Ghafoor**, K. Ismail, M. A. Rafiq. Symmetry driven second harmonic generation and tensor anisotropy in O-terminated MXenes. *Applied Physics A*, submitted and under review (2026).
- **F. Ghafoor**, M. N. Aslam, A. Rafiq, F. Siddique, K. Ismail. Dopant-Induced Modulation of the Electronic and Infrared Optical Properties of Zr_2CO_2 MXene. *Physica B: Condensed Matter*, in process (2026).
- K. Ismail, A. B. M. Ali, **F. Ghafoor**, et al. Electronic, optical and thermoelectric performance of Cs_2AuXY_6 for photovoltaic

Visiting Lecturer

COMWAVE Institute of Science and Information Technology

2009–2010 Islamabad, Pakistan

- Taught Optics and Modern Physics, Electromagnetic Theory, and Electronic Instrumentation.

Remote Graduate-Level Physics Instructor

University of Jyväskylä (JYU)

2022 Remote

- Delivered a Particle Physics course remotely to graduate students.

Lecturer in Physics

Government Degree College for Women

2010–2011 Taxila, Pakistan

- Delivered core physics instruction with structured problem solving, concept reinforcement, and examination preparation.

TEACHING AREAS

- General Physics / Introductory Physics
- Mechanics and Mathematical Methods
- Electricity and Magnetism
- Waves and Optics
- Thermal and Statistical Physics
- Modern Physics
- Quantum Mechanics
- Solid State / Condensed Matter Physics
- Computational Physics
- Materials Physics and Nanomaterials

COURSES TAUGHT

- Quantum Mechanics
- Advanced Mathematical Methods in Physics (Advanced MMP)
- Mathematical Methods in Physics (MMP)
- Atomic and Molecular Physics
- Particle Physics
- Advanced Particle Physics
- Calculus and Analytical Geometry (MS-level course)
- Advanced Computational Condensed Matter Physics
- Advanced Computational Materials / Advanced DMMP
- Materials Science
- Applied Physics

ACADEMIC LEADERSHIP & OUTREACH

- Member, FOWSD Pakistan Chapter (Forum for Women in Science Development).
- Disability Advisor and accessibility advocate, supporting inclusive academic participation and disability-aware educational practices.
- Physics advocate and mentor for Cambridge and Canadian curriculum learners.
- Capacity-building member supporting academic and educational development for Cambridge and Canadian courses.

applications: by DFT method. *Journal of Molecular Graphics and Modelling*, 142, 109203 (2025).

- A. Shakoor, B. ul Haq, S. Azam, **F. Ghafoor**, et al. DFT investigation of structural, electronic, optical, mechanical, and thermoelectric properties of halide perovskites $XZnCl_3$ (X = Ag, Au, Cu). *Physica B: Condensed Matter*, 712, 417309 (2025).
- **F. Ghafoor**, M. Ali, M. A. Rafiq, F. Siddique, A. Ali, A. Rauf. Modulating band gap and optical activity in GaN/Zr_2CO_2 heterostructure via stacking strategies for promising optoelectronic applications. *Computational Materials Science*, 251, 113727 (2025).
- **F. Ghafoor**, K. Ismail, M. A. Rafiq, M. Ali, F. Siddique. Quantum insights into O-terminated transition metal carbides: Structural and functional perspective for energy solutions. *Physica B: Condensed Matter*, 701, 416964 (2025).
- R. Khan, **F. Ghafoor**, Q. Zhang, et al. A first principles study of enhanced ferromagnetism in a two dimensional Cr doped InS monolayer. *Journal of Electronic Materials*, 51(11), 6252–6263 (2022).
- **F. Ghafoor**, M. Arfan, A. U. Rahman, M. A. Rafiq, E. S. Yousef. Enhanced electronic and thermoelectric properties of Zr_2CO_2/GaS van der Waals heterostructure: A first principles study. *Surfaces and Interfaces*, 35, 102453 (2022).
- **F. Ghafoor**. Top quark mass measurement at TEVATRON and LHC energies. *University of Wah Journal of Science and Technology*, 2, 31–34 (2018).

RESEARCH IN PROGRESS

Computational Optimization for Next-Generation Solar Evolution in Advanced Tin-Based Metal Halide Perovskites

- **Authors:** Fakhra Ghafoor, M. Naeem, M. Aftab Rafiq
- **Target Journal:** *AIP Advances*
- Computational investigation of advanced tin-based metal halide perovskites for next-generation solar-energy applications.

RESEARCH PROFILE

- DFT and TDDFT dynamical studies of two-dimensional (2D) materials and MXenes
- Linear and nonlinear optics, including first-principles optical-response and second-harmonic-generation studies
- Next-generation energy solutions, including photovoltaic, energy-conversion, and functional materials
- Infrared sensing devices and infrared optical response of functional materials
- Perovskites and advanced photovoltaic materials
- Bulk materials and semiconductor heterostructures
- Thermoelectric materials and transport properties
- First-principles materials modelling, scientific computing, and dynamical response calculations

ACADEMIC ACTIVITIES & DISTINCTIONS

- **Speaker, ISoAD 2026, April 2026** — research presentation at an international academic event.
- **Best Poster Award, ISPAD 2024, April 2024** — awarded for poster presentation at the International School on Physics & Allied Disciplines, jointly organized by NCP and ICTP Italy.

COMPUTATIONAL SKILLS

- Density Functional Theory (DFT), first-principles calculations, electronic structure and optical-property analysis.
- Python, C, C++, MATLAB, Bash and LaTeX.
- Linux-based scientific computing and reproducible computational workflows.
- Quantum ESPRESSO, WIEN2k, VASP, EXCITING, VESTA and XCrySDen.

REFERENCES

Prof. Dr. Muhammad Aftab Rafiq
Director General, National Centre for Physics (NCP)
Islamabad, Pakistan

Prof. Dr. Uzma Ghazanfar
Dean, Faculty of Basic Sciences
University of Wah, Wah Cantt, Pakistan

Prof. Dr. Ahmad Sohail
Associate Professor, Department of Mechanical Engineering
College of Electrical & Mechanical Engineering (CE&ME), NUST
Islamabad, Pakistan

ONLINE ACADEMIC PROFILE

Google Scholar Profile
LinkedIn Profile

- **Remote Graduate-Level Teaching: Particle Physics, University of Jyväskylä (JYU)** — graduate-level teaching experience.
- Speaker, LHC School organized by National Centre for Physics and ICTP Italy.
- Participant, International Nathiagali Summer College on Physics and Contemporary Needs.
- Participant, MAX School on Advanced Materials and Molecular Modeling with Quantum ESPRESSO.
- Participant, Workshop on 2D Materials for Spin Orbitronics.
- Participant, National Workshop on Physics Education, Challenges and Prospects, National Centre for Physics, QAU.
- Workshop on Preparation of Self Assessment Report, University of Wah.